1.Write a Program that accepts two Strings as command line arguments and generate the output in the required format.
Example1) If the two command line arguments are Wipro and Bangalore then the output generated should be Wipro Technologies Bangalore.

```java
public class Main {
    public static void main(String[] args) {
        String str = "Technologies ";
        System.out.println(args[0] +" "+ str + args[1]);
    }
}
```

Main

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ
Wipro Technologies Bangalore

Process finished with exit code 0
```

2.Write a Program to accept a String as a command line argument and print a Welcome message as given below. Example1) C:\> java Sample John O/P Expected : Welcome John

```java
public class Main {
    public static void main(String[] args) {
        String str = " Balaji";
        System.out.println(args[0]+str);
    }
}
```

Main

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
Welcome Balaji

Process finished with exit code 0
```

3.Write a Program to accept two integers as command line arguments and print the sum of the two numbers Example1) C:\>java Sample 10 20
O/P Expected : The sum of 10 and 20 is 30

```java
public class Main {
    public static void main(String[] args) {
        int num1 = Integer.parseInt(args[0]);
        int num2 = Integer.parseInt(args[1]);
        System.out.println(" The sum of "+args[0]+" and "+args[1]+" is "+(num1+num2));
    }
}
```

Main

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
 The sum of 10 and 20 is 30

Process finished with exit code 0
```

4.Write a program to check if a given integer number is Positive, Negative, or Zero.

```java
public class Practiceprob {
    public static void main(String[] args){
        int n = -5;
        if(n<0){
            System.out.println("Negative");
        }
        else if(n>0){
            System.out.println("Positive");
        }
        else{
            System.out.println("Zero");
        }
    }
}
```

Practiceprob

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
Negative

Process finished with exit code 0
```

5.Write a program to check if a given integer number is odd or even.

```java
public class Practiceprob {
    public static void main(String[] args){
        int n = 4;
        if(n%2==0){
            System.out.println(n+" is even");
        }
        else{
            System.out.println(n+" is odd");
        }
    }
}
```

Practiceprob

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
4 is even

Process finished with exit code 0
```

6.Write a program to check if the program has received command line arguments or not. If the program has not received arguments then print "No Values", else print all the values in a single line separated by ,(comma) Example1) java Example O/P: No values Example2) java Example Mumbai Bangalore O/P: Mumbai,Bangalore

```java
public class Main {
    public static void main(String[] args) {
        int n = args.length;
        if(n==0) System.out.println("No values");
        for(int i=0;i<n;i++){
            if(i!=n-1){
                System.out.print(args[i]+",");
            }
            else{
                System.out.print(args[i]);
            }
        }
    }
}
```

Main

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ
Mumbai,Bangalore,Chennai
Process finished with exit code 0
```

7.Initialize two character variables in a program and display the characters in alphabetical order. Example1) if the first character is 's' and second character is 'e' then the output should be e,s Example2) if the first character is 'a' and second character is 'e', then the output should be a,e

```java
public class Practiceprob {
    public static void main(String[] args){
        char a = 'e';
        char b = 'a';
        if(a>b){
            System.out.println(a+" "+b);
        }
        else{
            System.out.println(b+" "+a);
        }
    }
}
```

Practiceprob

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
e a

Process finished with exit code 0
```

8.Initialize a character variable in a program and print 'Alphabhet' if the initialized value is an alphabhet, print 'Digit' if the initialized value is a number, and print 'Special Character', if the initialized value is anything else.

```java
public class Practiceprob {
    public static void main(String[] args){
        char a = '@';
        if((a>='a' && a<='z') || (a>='A' && a<='Z')){
            System.out.println("Alphabhet");
        }
        else if(a>='0' && a<='9'){
            System.out.println("Digit");
        }
        else{
            System.out.println("Special Characters");
        }
    }
}
```

Practiceprob

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
Special Characters

Process finished with exit code 0
```

9.Write a program to accept gender ("Male" or "Female") and age from command line arguments and print the percentage of interest based on the given conditions. If the gender is 'Female' and age is between 1 and 58, the percentage of interest is 8.2%. If the gender is 'Female' and age is between 59 and 100, the percentage of interest is 9.2%. If Statement If the gender is 'Male' and age is between 1 and 58, the percentage of interest is 8.4%. If the gender is 'Male' and age is between 59 and 100, the percentage of interest is 10.5%.

```java
public class Main {
    public static void main(String[] args) {
        String gender = "Male";
        int age = Integer.parseInt(args[0]);
        if(gender.equals("Male")){
            if(age>=1 && age<=58) System.out.println("8.4%");
            else System.out.println("10.5");
        }
        else{
            if(age>=1 && age<=58) System.out.println("8.2%");
            else System.out.println("9.2");
        }
    }
}
```

Main

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ
8.4%

Process finished with exit code 0
```

10.Initialize a character variable with an alphabhet in a program. If the character value is in lowercase, the output should be displayed in uppercase in the following format. Example1) i/p:a o/p:a->A If the character value is in uppercase, the output should be displayed in lowercase in the following format. Example2) i/p:A o/p:A->a

```java
public class Main {
    public static void main(String[] args) {
        char c = 'A';
        if(c>='a' && c<='z'){
            int val = c-32;
            char c1 = (char)val;
            System.out.println(c+"->"+c1);
        }
        else{
            int val1 = c+32;
            char c2  = (char)val1;
            System.out.println(c+"->"+c2);
        }
    }
}
```

Main

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ
A->a

Process finished with exit code 0
```

11.Write a program to receive a color code from the user (an Alphabhet). The program should then print the color name, based on the color code given. The following are the color codes and their corresponding color names. R->Red, B->Blue, G->Green, O->Orange, Y->Yellow, W >White. If color code provided by the user is not valid then print "Invalid Code".

```java
public class Main {
    public static void main(String[] args) {
        char colorCode = 'B';
        switch(colorCode){
            case 'R':
                System.out.println("Red");
                break;
            case 'B':
                System.out.println("Blue");
                break;
            case 'G':
                System.out.println("Green");
                break;
            case 'O':
                System.out.println("Orange");
                break;
            case 'Y':
                System.out.println("Yellow");
                break;
            case 'W':
                System.out.println("White");
                break;
            default:
                System.out.println("Invalid Code");
```

Main

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community
Blue

Process finished with exit code 0
```

Write a program to receive a number and print the corresponding month name. Example1) C:\>java Sample 12 O/P Expected : December Example2) C:\>java Sample O/P Expected : Please enter the month in numbers Example3) C:\>java Sample 15 O/P Expected : Invalid month

HomeControll... | Expense.java | ExpenseCont... | application... | Main.java

```java
package main;

public class Main {
    public static void main(String[] args) {
        int month =  Integer.parseInt(args[0]) ;
        switch(month) {
           case 1:
               System.out.print("January");
               break;
           case 2:
               System.out.print("Febuary");
               break;
           case 3:
               System.out.print("March");
               break;
           case 4:
               System.out.print("April");
               break;
           case 5:
               System.out.print("May");
               break;
           case 6:
               System.out.print("June");
               break;
           case 7:
               System.out.print("July");
               break;
           case 8:
               System.out.print("August");
               break;
           case 9:
               System.out.print("September");
               break;
           case 10:
               System.out.print("October");
               break;
           case 11:
               System.out.print("November");
               break;
           case 12:
               System.out.print("December");
               break;
```

Problems | Javadoc | Declaration | Console | Coverage

<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe

```
July
```

13.Write a program to print numbers from 1 to 10 in a single row with one tab space.

```java
public class Practiceprob {
    public static void main(String[] args){
        for(int i=1;i<=10;i++){
            System.out.print(i+" ");
        }
    }
}
```

Practiceprob

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program
1 2 3 4 5 6 7 8 9 10
Process finished with exit code 0
```

14.Write a program to print even numbers between 23 and 57. Each number should be printed in a separate row.

HomeControll... | Expense.java | ExpenseCont... | application... | Main.java

```java
package main;

public class Main {
    public static void main(String[] args) {
        for(int i=23;i<=57;i++){
            if(i%2==0){
                System.out.println(i);
            }
        }
    }
}
```

Problems | Javadoc | Declaration | Console | Coverage

<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (

```
24
26
28
30
32
34
36
38
40
42
44
46
48
50
52
54
56
```

15.Write a program to check if a given number is prime or not.

HomeControll... | Expense.java | ExpenseCont... | application... | Main.java

```java
package main;

public class Main {
    public static void main(String[] args) {
        int n = 97;
        boolean flag = true;
        if(n<=1) {
            System.out.println("Not Prime");
        }
        for(int i=2;i*i<=n;i++) {
            if(n%i==0) {
                flag = false;
                System.out.println("Not Prime");
            }
        }
        if(flag) {
            System.out.println("Prime");
        }
    }
}
```

Problems | Javadoc | Declaration | Console | Coverage

<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (

```
Prime
```

16.Write a program to print prime numbers between 10 and 99.

HomeControll... | Expense.java | ExpenseCont... | application... | Main.java

```java
package main;

public class Main {
    public static void main(String[] args) {
        for(int i=10;i<=99;i++) {
            boolean flag = helperMethod(i);
            if(flag) {
                System.out.println(i);
            }
        }
    }
    public static boolean helperMethod(int n) {
        if(n<=1){
            return false;
        }
        boolean flag = true;
        for(int i=2;i*i<=n;i++) {
            if(n%i==0) {
                flag = false;
                return false;
            }
        }
        return flag;
    }
}
```

Problems | Javadoc | Declaration | Console | Coverage

<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (

```
11
13
17
19
23
29
31
37
41
43
47
53
59
61
67
71
73
79
83
89
97
```

17.Write a Java program to find if the given number is prime or not. Example1) C:\>java Sample O/P: Please enter an integer number Example2) C:\>java Sample 1 O/P:1 is neither prime nor composite Example3) C:\>java Sample 0 O/P: 0 is neither prime nor composite Example4) C:\>java Sample 10 O/P: 10 is not a prime number Example5) C:\>java Sample 7 O/P : 7 is a prime number

HomeControll... | Expense.java | ExpenseCont... | application... | Main.java

```java
package main;

public class Main {
    public static void main(String[] args) {
        int n = Integer.parseInt(args[0]);
        boolean flag = true;
        if(n<=1) {
            System.out.println(n+" is neither prime nor composite");
        }
        for(int i=2;i*i<=n;i++) {
            if(n%i==0) {
                flag = false;
                System.out.println(n+" is a Not Prime");
            }
        }
        if(flag) {
            System.out.println(n+" is a Prime");
        }
    }
}
```

Problems | Javadoc | Declaration | Console | Coverage

<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (

```
97 is a Prime
```

18.Write a program to print the sum of all the digits of a given number. Example1) I/P:1234 O/P:10

```java
package main;

public class Main {
    public static void main(String[] args) {
        int n = 1234;
        int sum = 0;
        for(int i=n;i>0;i /= 10) {
            sum += i%10;
        }
        System.out.println(sum);
    }
}
```

```
<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe
10
```

19.Write a program to print * in Floyds format (using for and while loop) * * * * * *

```java
package main;

public class Main {
    public static void main(String[] args) {
        int n = Integer.parseInt(args[0]);
        for(int i=0;i<n;i++) {
            for(int j=0;j<=i;j++) {
                System.out.print("*"+" ");
            }
            System.out.println();
        }
    }
}
```

```
<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (
*
* *
* * *
```

```java
package main;

public class Main {
    public static void main(String[] args) {
        int n = Integer.parseInt(args[0]);
        int i = 1;
        int j = 1;
        while(i<=n) {
            while(j<=i) {
                System.out.print("*"+" ");
                j++;
            }
            j = 1;
            System.out.println();
            i++;
        }
    }
}
```

```
<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (
*
* *
* * *
```

20.Write a program to reverse a given number and print Example1) I/P: 1234 O/P:4321 Example2) I/P:1004 O/P:4001

HomeControll... | Expense.java | ExpenseCont... | application... | Main.java

```java
package main;

public class Main {
    public static void main(String[] args) {
        int n = 1234;
        int reverse = 0;
        while(n!=0) {
            int digit = n%10;
            reverse = reverse*10+digit;
            n /= 10;
        }
        System.out.println(reverse);
    }
}
```

Problems | Javadoc | Declaration | Console | Coverage

<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (

```
4321
```

21.Write a Java program to find if the given number is palindrome or not Example1) C:\>java Sample 110011 O/P: 110011 is a palindrome Example2) C:\>java Sample 1234 O/P: 1234 is not a palindrome

HomeControll... | Expense.java | ExpenseCont... | application... | Main.java

```java
package main;

public class Main {
    public static void main(String[] args) {
        int n = Integer.parseInt(args[0]);
        int temp = n;
        int reverse = 0;
        while(temp!=0) {
            int digit = temp%10;
            reverse = reverse*10+digit;
            temp /= 10;
        }
        if(reverse==n) {
            System.out.println(n+" is a Palindrome");
        }
        else {
            System.out.println(n+" is not Palindrome");
        }
    }
}
```

Problems | Javadoc | Declaration | Console | Coverage

```
<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (
110011 is a Palindrome
```

22.Write a program to print first 5 values which are divisible by 2, 3, and 5.

```java
package main;

public class Main {
    public static void main(String[] args) {
        int i = 1;
        int count = 0;
        while(count!=5) {
            if(i%2==0 && i%3==0 && i%5==0){
                System.out.print(i+" ");
                count++;
            }
            i++;
        }
    }
}
```

```
<terminated> Main [Java Application] C:\Program Files\Java\jdk-23\bin\javaw.exe (
30 60 90 120 150
```